PCBC H Parametrics Practice Test A Name _____ _____/12

Calculator | 30 minutes total for all four questions

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

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Signature _____

SHOW ALL WORK FOR FULL CREDIT!

Practice set: the calculator question comes first, as announced in class. Every question copies a homework problem with new numbers. 3 points each is a practice split of the posted 12.

Show your set up (what you are putting into the calculator) and round your answers to the nearest thousandth. *(practice instructions)*

1. The path of a projectile follows the parametric equations

$$x = v_0(\cos\theta)t \qquad \text{and} \qquad y = -16t^2 + v_0(\sin\theta)t + h_0$$

where θ is the angle of inclination of the projectile at the launch, v_0 is the initial velocity of the projectile in feet per second, and h_0 is the initial height of the projectile in feet.

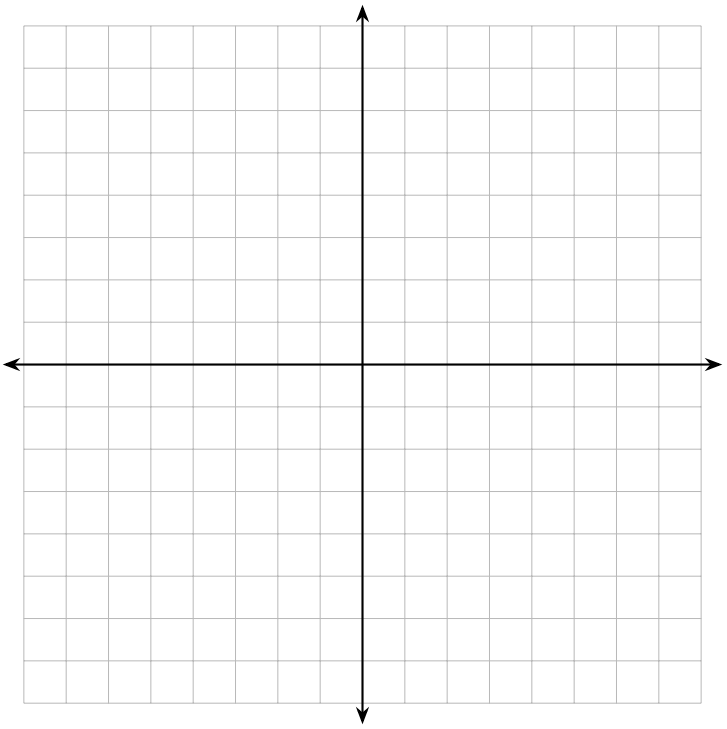
A baseball is hit from a height of 3 ft at an angle of inclination of 35° with a velocity of 110 ft/sec. (a) Write the parametric equations that model its path. (b) For how many seconds is the baseball in flight, and what horizontal distance does it travel? **3 pts**

Non-Calculator | 30 minutes total for all four questions

SHOW ALL WORK FOR FULL CREDIT!

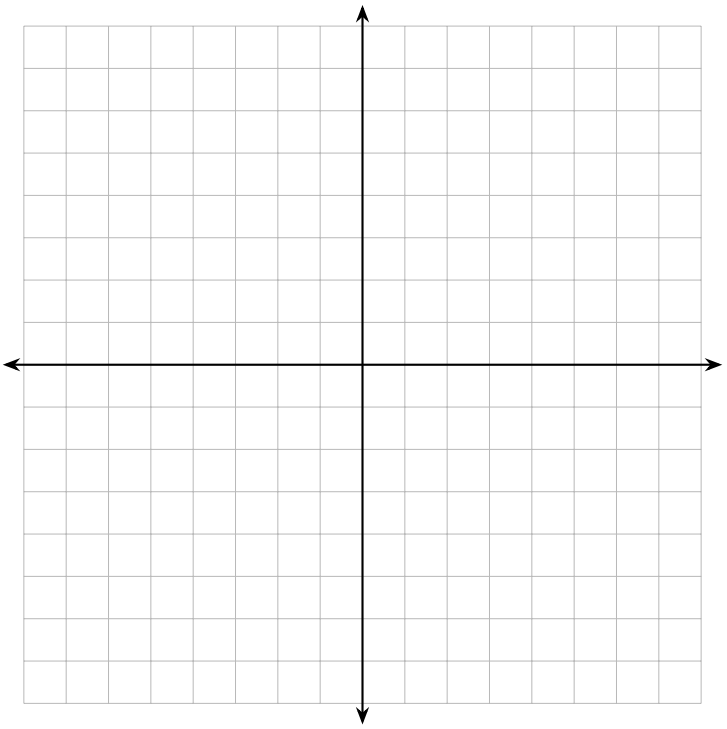
(a) Find the point that corresponds to $t = 0$. (b) Graph the plane curve defined by the given parametric equations. Then find a rectangular equation for the curve. State any restrictions on the domain and show the direction of the curve. (3 points)

2. $x = \sqrt{t+1}, \quad y = 2t, \quad$ for t in $[-1, 3]$ **3 pts**



Find a rectangular equation for the given plane curve, and graph the curve. Show the direction of the curve. (3 points)

3. $x = 4\sin t, \quad y = 2\cos t, \quad 0 \le t \le 2\pi$ **3 pts**



Find two different parametric representations for the equation of the parabola. (3 points)

4. $y = x^2 - 4x + 1$ **3 pts**

Representation 1 *Representation 2*

PCBC H Parametrics Practice Test A: Answer Key

12 points

Each solution follows the method used in class; the gray box shows another way. If the question does not say *simplify*, any form with t gone is fine (a form that shows the graph is best). *Simplify* means get y alone. Direction arrows only when the question asks. The calculator question allows small rounding differences. Every value was checked in Python.

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1. *Calculator* (like homework day 2 #20)

(a) $\boxed{x = (110\cos 35^\circ)t, \quad y = -16t^2 + (110\sin 35^\circ)t + 3}$

(b) in flight $\boxed{\approx 3.990 \text{ sec}}$, travels $\boxed{\approx 359.555 \text{ ft}}$

In class	Why
$x = (110\cos 35^\circ)t$	$v_0 \cos\theta$ times t. Keep the degree symbol.
$y = -16t^2 + (110\sin 35^\circ)t + 3$	Starting height goes on the end.
Degree mode. Graphs, Parametric: $x1(t) = 110\cos(35)t$, $y1(t) = -16t^2 + 110\sin(35)t + 3$	Enter it exactly as written.
t from 0 to 5	If the path stops before the ground, raise the t max (it starts at 2π).
Trace to where $y = 0$; set the trace step to 0.01, then 0.001, until y is as close to 0 as you can get	Precision to the thousandths.
Read $t \approx 3.990$ and $x \approx 359.555$	t is the time in flight, x the distance.

Other way. Set $y = 0$: $0 = -16t^2 + (110\sin 35^\circ)t + 3$. Quadratic formula gives $t \approx 3.990$ (reject the negative). Then $x = (110\cos 35^\circ)(3.99033\ldots) \approx 359.555$. Use the unrounded t.

Practice scoring (3 pts): 1 for the equations, 1 for the time, 1 for the distance.

Trap: radian mode, or rounding t to 3.99 before finding x (≈ 359.526).

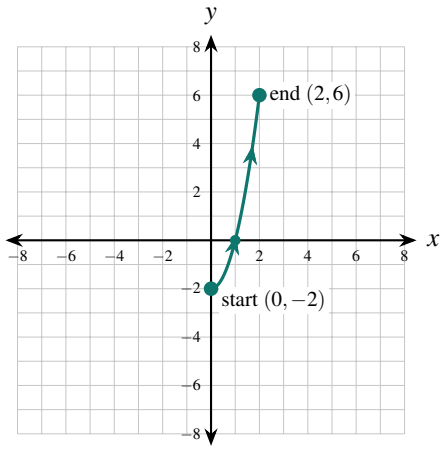
2. $x = \sqrt{t+1}, \quad y = 2t, \quad$ for t in $[-1,3]$ *Non-Calculator* (like homework day 1 #1, #2 and #4)

(a) $\boxed{(1,0)}$ (b) $\boxed{y = 2x^2 - 2, \text{ for } x \text{ in } [0,2]}$
Starts at $(0,-2)$, ends at $(2,6)$, solid dots, no arrows on the ends.

In class	Why
(a) $x = \sqrt{0+1} = 1,\ y = 2(0) = 0$	Plug $t = 0$ into both.
t-chart: $t = -1, 0, 1, 2, 3$	Brackets: both ends go in the chart.
Plot, connect, arrows from $t = -1$ to $t = 3$	Lowest t to highest t.
$x^2 = t + 1$, so $t = x^2 - 1$	Get t alone (one answer, no $\pm$).
$y = 2(x^2 - 1)$	Substitute into y. Distributing to $2x^2 - 2$ is optional.
for x in $[0,2]$	Min and max of the x column.

t	(x,y)
-1	$(0,-2)$
0	$(1,0)$
1	$(\sqrt{2},2)$
2	$(\sqrt{3},4)$
3	$(2,6)$

Closed interval, so solid dots at both ends and no arrows on the ends.



Other way. Isolate t from y instead: $t = \frac{y}{2}$, so $x = \sqrt{\frac{y}{2} + 1}$ for x in $[0,2]$. Same curve, x alone.

Practice scoring (3 pts): 1 for $(1,0)$ and the equation, 1 for x in $[0,2]$, 1 for the graph with end dots and direction.
Trap: drawing the whole parabola. $\sqrt{\ }$ is never negative and t stops at 3.

3. $x = 4\sin t, \quad y = 2\cos t, \quad 0 \le t \le 2\pi$ *Non-Calculator* (like homework day 1 #14)

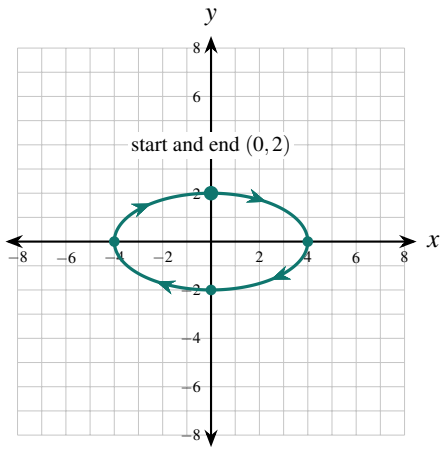
$\boxed{\dfrac{x^2}{16} + \dfrac{y^2}{4} = 1}$

Whole ellipse, traced clockwise, starting and ending at $(0,2)$. No domain needed.

In class	**Why**
t-chart: $0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$	Quadrant values.
Plot, connect, arrows: $(0,2)$ to $(4,0)$	Top, then right: clockwise.
$\sin t = \dfrac{x}{4}, \quad \cos t = \dfrac{y}{2}$	Can't get t alone, so isolate the trig.
$\dfrac{x^2}{16} = \sin^2 t, \quad \dfrac{y^2}{4} = \cos^2 t$	Square both sides.
$\dfrac{x^2}{16} + \dfrac{y^2}{4} = 1$	$\sin^2 t + \cos^2 t = 1$.

t	(x,y)
0	$(0,2)$
$\frac{\pi}{2}$	$(4,0)$
π	$(0,-2)$
$\frac{3\pi}{2}$	$(-4,0)$
2π	$(0,2)$

An ellipse restricts itself, so no "for x in" is needed.



Other way. Read it off the numbers: $4\sin t$ means 4 left and right, $2\cos t$ means 2 up and down, center $(0,0)$. Plug in $t = 0$ and $t = \frac{\pi}{2}$ for the direction.

Practice scoring (3 pts): 1 for the equation, 1 for the ellipse drawn to size, 1 for the clockwise direction.

Trap: assuming counterclockwise. sin in x and cos in y starts at the top and goes clockwise.

4. $y = x^2 - 4x + 1$ *Non-Calculator* (like homework day 1 #16)

$\boxed{x = t,\ y = t^2 - 4t + 1,\ \text{for } t \text{ in } (-\infty, \infty)}$ and $\boxed{x = t + 2,\ y = t^2 - 3,\ \text{for } t \text{ in } (-\infty, \infty)}$

In class	Why
$x = t$, so $y = t^2 - 4t + 1$	Representation 1: let x be t.
$y = (x^2 - 4x + 4) - 4 + 1 = (x - 2)^2 - 3$	Complete the square.
$t = x - 2$, so $x = t + 2$	Let t be the inside.
$y = t^2 - 3$	Representation 2.
for t in $(-\infty, \infty)$ on both	Every x is allowed.

Other way. Any substitution that covers every x works, for example $x = 2 - t$: $y = (2 - t)^2 - 4(2 - t) + 1 = t^2 - 3$.

Practice scoring (3 pts): 1 for each correct pair, 1 for the t interval on both.

Trap: $t = x - 2$ means $x = t + 2$, not $t - 2$.